Answer correction

1.57

1.57 Determine whether the following signals are periodic, and for those which are, find the fundamental period:

(a) $x[n] = \cos\left(\frac{8}{15}\pi n\right)$

(b) $x[n] = \cos\left(\frac{7}{15}\pi n\right)$

$\boxed{\text{(c)}\ x(t) = \cos(2t) + \sin(3t)}$

(d) $x(t) = \sum_{k=-\infty}^{\infty}(-1)^k\delta(t - 2k)$

(e) $x[n] = \sum_{k=-\infty}^{\infty}\{\delta[n - 3k] + \delta[n - k^2]\}$

(f) $x(t) = \cos(t)u(t)$

(g) $x(t) = v(t) + v(-t)$, where $v(t) = \cos(t)u(t)$

(h) $x(t) = v(t) + v(-t)$, where $v(t) = \sin(t)u(t)$

(i) $x[n] = \cos\left(\frac{1}{5}\pi n\right)\sin\left(\frac{1}{3}\pi n\right)$

(c) **Periodic**

Fundamental period = 2π samples

1.64

1.64 The systems that follow have input $x(t)$ or $x[n]$ and output $y(t)$ or $y[n]$. For each system, determine whether it is (i) memoryless, (ii) stable, (iii) causal, (iv) linear, and (v) time invariant.

(a) $y(t) = \cos(x(t))$

(b) $y[n] = 2x[n]u[n]$

(c) $y[n] = \log_{10}(|x[n]|)$

(d) $y(t) = \int_{-\infty}^{t/2} x(\tau)\, d\tau$

(e) $y[n] = \sum_{k=-\infty}^{n} x[k + 2]$

(f) $y(t) = \frac{d}{dt}x(t)$

(g) $y[n] = \cos(2\pi x[n + 1]) + x[n]$

(h) $y(t) = \frac{d}{dt}\{e^{-t}x(t)\}$

$\boxed{\text{(i)}\ y(t) = x(2 - t)}$

(j) $y[n] = x[n]\sum_{k=-\infty}^{\infty}\delta[n - 2k]$

$\boxed{\text{(k)}\ y(t) = x(t/2)}$

(l) $y[n] = 2x[2^n]$

(i) and (k) are both **time-variant**.